

Excerpt of PPGface: Like What You Are Watching? Earphones Can “Feel” Your Facial Expressions

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ABSTRACT

Facial expression recognition has been widely explored to demonstrate people’s emotional states. However, existing systems primarily rely on external devices which seems less accessible and efficient. To this end, we propose PPGface, a ubiquitous facial expression recognition platform that leverages earable devices with built-in PPG sensor. PPGface understands the facial expressions through the dynamic PPG patterns resulting from facial muscle movements. Through several comprehensive studies, this work validates a great potential to be employed in future commodity earable devices.

CCS CONCEPTS

• **Human-centered computing** → **Ubiquitous and mobile devices**; **Human computer interaction (HCI)**.

KEYWORDS

Photoplethysmogram, PPG, Facial Expression, Ear Canal, Blood Vessel Deformation

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1 INTRODUCTION

As more viewers are moving from traditional pay-TV to streaming market trend, many media and service providers have explored

different approaches to capture users’ emotional reactions on the fly through video watching habits [2]. Facial expressions have been deemed as the universal non-verbal language to express internal emotional states, which are also accompanied by unconscious body postures. Although there are few approaches to understand facial expressions, they either raise privacy issues (computer vision-based) or require external hardware setup close to the target user (WiFi-based). To enable a more privacy-preserving and ubiquitous system, we propose PPGface, which leverages in-ear PPG signals with complementary accelerometer data. Specifically, PPGface understands different facial expressions through dynamic PPG patterns which are altered by the facial muscle motions where the signals are collected inside the ear.

2 RELATED WORK

Computer vision (CV) based Facial expression recognition (FER) system is one of the most intuitive way to recognize different facial expressions. Zeng *et al.* [5] proposed the deep sparse autoencoders (DSAE) which extracts meaningful features in an unsupervised manner.

Besides the CV-based approach, researchers employed different sensing modalities to build FER system. For example, Chen *et al.* [1] introduced a WiFi based facial expression recognition system to classify six different emotions by using a laptop and three antennas. In addition, Gruebler *et al.* [3] proposes a wearable device embedded with an EMG sensor and shows that the device can continuously track smiling and frowning. Although some perform relatively good, it is unrealistic to setup extra hardware devices while tracking facial expressions or impractical to attach multiple sensors to the face in daily lives.

Researchers have employed PPG signals to measure cardiac activities such as heart rate or blood pressure. Due to the fact that PPG signals are also vulnerable to motion artifacts, previous studies also focused on removing those motion artifacts. Recently, unlike previous research objectives, studies have started to re-purpose the PPG motion artifacts instead of regarding them as noises. Zhao *et al.* [6] proposed a gesture recognition system using PPG signals and motion sensors embedded in a wrist worn device that can discriminate finger movements.

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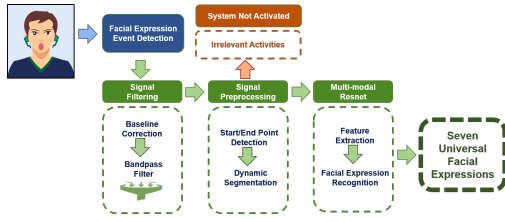


Figure 1: PPGface system overview

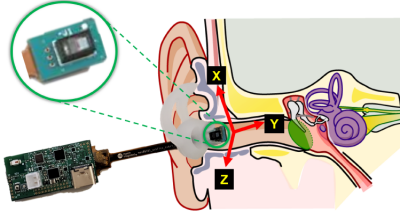


Figure 2: The depiction of the sensor

3 PPGFACE SYSTEM DESIGN

As shown in Fig. 1, our proposed FER system, PPGface, is to understand the user's different facial expressions by fusing both the behavioral (IMU signals from spontaneous body posture) and physiological (PPG signals from the facial muscle movement) patterns.

First, the device measures the in-ear PPG and accelerometer signals when user makes an expression. The measured signals show distinctive patterns between the resting stage and when the user makes a facial expression. In addition, if the user performs other physical activities which are irrelevant to facial expressions, such as yawning or swallowing saliva, these unrelated signals are detected and removed for the further steps.

After the event detection step, we apply detrend technique to remove the DC components caused by respiration followed by the bandpass filter, which filters out both the high and low frequency components to preserve the characteristics of both the facial expressions and related spontaneous body postures.

The dynamic signal preprocessing step involves two sub-stages: segmentation and detection. First, the peak detection algorithm and continuous wavelet transform (CWT) technique are used to select possible candidate sets of start and end points of the PPGface signals. By analyzing these signals, the system is able to detect the facial expression activity events, given the final start and end points of each segment corresponding to the facial expression activities.

At last, preprocessed signals will be the inputs to the classification module. To overcome the limited amount of data, we apply the following data augmentation methods: Rotation, time-warping, and scaling techniques. Then, multimodal ResNet is used to extract the representative features and classify seven different facial expressions. The user's profile is generated to classify seven facial expressions during the training stage. During the testing, PPGface goes through feature profiling to classify any unseen data based on the pre-train model.

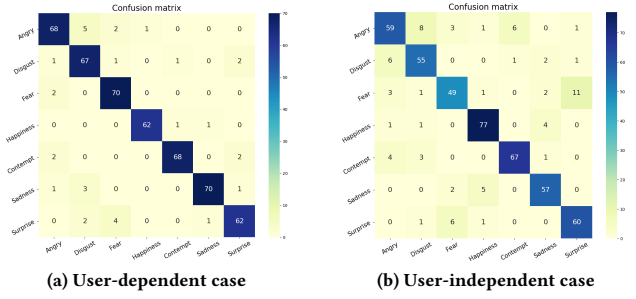


Figure 3: Confusion matrix of seven universal facial expressions. (a) user-dependent, (b) user-independent case.

4 EXPERIMENTAL SETUP AND RESULTS

In this study, we deployed MAXIM86161EVSYS evaluation kit [4] as shown in Fig. 2. The sampling rate of three LDEs (i.e., green, infrared (IR), and red) was set to 128 Hz. Along with the LEDs, the sensor is also equipped with 3-axis accelerometer sensors synchronized with PPG sensor.

We evaluate the user-dependent performance of the PPGface in terms of classification accuracy. PPGface achieves 0.935 accuracy (STD = 0.086), 0.951 precision (STD = 0.086), 0.934 recall (STD = 0.074), and 0.934 f1-score (STD = 0.094) (STD: Standard Deviation). It is observed from Fig. 3a that accuracy of fear and surprise is lower than other expressions, which is because facial muscles to express those expressions overlap a lot.

We wanted to explore more whether PPGface is also able to distinguish well under the user-independent case. In this case, we combined each facial expression without considering the user label. Performance of the PPGface's accuracy is 0.848, precision is 0.854 (STD = 0.048), recall is 0.854 (STD = 0.074), and f1-score is 0.842 (STD = 0.054). Performance shows lower accuracy than the user-dependent case due to a difference among each user when making facial expressions which can be shown from Fig. 3b.

5 CONCLUSION

We introduce PPGface, a novel approach to understand different facial expressions by leveraging unique in-ear PPG variations with the aid of an accelerometer. It is believed that this work has a potential to provide insight into non-intrusive in-ear sensing research. In the future, we plan to design customized prototype, enhance the versatility of PPGface, and examine the permanence and robustness from a larger and more diverse subject pool.

REFERENCES

- [1] Yanjiao Chen, Runmin Ou, Zhiyang Li, and Kaishun Wu. 2020. WiFace: facial expression recognition using Wi-Fi signals. *IEEE Transactions on Mobile Computing* (2020).
- [2] Michael Goodman. 2021. *U.S. SVOD Forecast (2010 - 2026)*. Technical Report. Strategy Analytics, Newton, Massachusetts USA. <https://www.strategyanalytics.com/> Last accessed: 2021-08-25.
- [3] Anna Gruebler and Kenji Suzuki. 2010. Measurement of distal EMG signals using a wearable device for reading facial expressions. In *Proceedings of the 2010 Annual International Conference of the IEEE Engineering in Medicine and Biology*. IEEE, 4594–4597.
- [4] Maxim Integrated. 2021. MAXM86161 Single-Supply Integrated Optical Module for HR and SpO2 Measurement. Data Sheet.

- [5] Nianyin Zeng, Hong Zhang, Baoye Song, Weibo Liu, Yurong Li, and Abdullah M Dobaie. 2018. Facial expression recognition via learning deep sparse autoencoders. *Neurocomputing* 273 (2018), 643–649.
- [6] Tianming Zhao, Jian Liu, Yan Wang, Hongbo Liu, and Yingying Chen. 2019. Towards Low-cost Sign Language Gesture Recognition Leveraging Wearables. *IEEE Transactions on Mobile Computing* 20, 4 (2019), 1685–1701.